Powder Technology 413 (2023) 117998

Contents lists available at ScienceDirect

Powder Technology

journal homepage: www.journals.elsevier.com/powder-technology

Modeling gravity filling of dies on a rotary tablet press

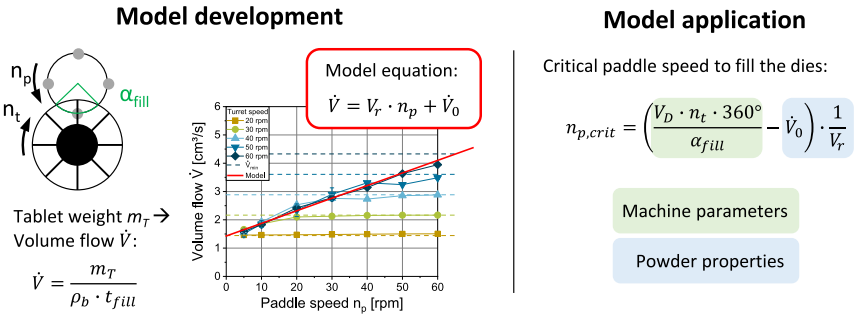
A.K. Schomberg [a,b,*], A. Kwade [a,b], J.H. Finke [a,b]

[a] Institute for Particle Technology, Technische Universität Braunschweig, Volkmaroder Str.5, 38104 Braunschweig, Germany
[b] Center of Pharmaceutical Engineering (PVZ), Technische Universität Braunschweig, Franz-Liszt-Str. 35A, 38106 Braunschweig, Germany

HIGHLIGHTS

- Investigation of the gravity filling process on a rotary tablet press in pilot-scale.
- Development of model to describe gravity filling process.
- Correlation of model parameters with powder properties like permeability and bulk density.
- Prediction of the minimal paddle speed to fill dies completely for a given turret speed by using the model.
- Very good agreement between predicted and experimentally determined minimal paddle speed for all investigated materials.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Die filling
Rotary tablet press
Gravity filling
Modeling

ABSTRACT

For tableting on rotary presses, die filling is the qualitydominating process step as the tablet weight and its variation is influenced. Therefore, a deeper understanding of this process is necessary. In this study, the filled powder weight was investigated for various parameter combinations on a rotary tablet press in pilot-scale. Based on the obtained data for four materials, which differ in density and flow properties, a process-property model was developed to describe the volume flow into the dies. The flow can be used to calculate either the final tablet weight or the minimal paddle speed to fill the dies completely at a set turret speed, which supports process development. Finally, model parameters were correlated with the materials bulk density and permeability. Therefore, this models enables the prediction of the die filling process only based on geometrical data of the rotary press, process parameters and measureable material properties, for the first time.

1. Introduction

Die filling is the crucial step for meeting the most critical quality attributes during tableting on rotary presses as it affects not only the filled powder weight and weight uniformity but also the resulting tablet porosity and tensile strength and, consecutively, the disintegration and dissolution behavior [1]. Effects of die filling on the tablet porosity and tensile strength result from a constant minimal punch distance during compaction on rotary presses, which leads to varying compression stresses if the powder weight fluctuates. Commonly, the choice of process parameters for the production on rotary tablet presses is done empirically and, by that, often results in non-optimal process parameter combinations. In worst case, the incomplete and varying die filling results in tablet properties which fail to meet the critical quality

* Corresponding author at: Institute for Particle Technology, Technische Universität Braunschweig, Volkmaroder Str.5, 38104 Braunschweig, Germany.
E-mail address: a.schomberg@tu-braunschweig.de (A.K. Schomberg).

https://doi.org/10.1016/j.powtec.2022.117998
Received 9 August 2022; Received in revised form 3 October 2022; Accepted 5 October 2022
Available online 14 October 2022
0032-5910/© 2022 Elsevier B.V. All rights reserved.

requirements and produced tablets have to be rejected. In order to avoid this, the paddle speed is often increased to enhance the amount of available powder above the dies during their feed frame passage and, thus, the filled powder weight [2–5]. However, paddle rotation induces shear stresses into the powder mixture which can result in unwanted changes of the powder properties, e.g. by abrasion or overlubrication [6–9]. The latter describes the enhanced lubricant dispersion within the formulation which possibly results in a considerable loss of compactability of the final tablets. Beside these scenarios, which have negative effects on the quality attributes, the maximum production rate for a given paddle speed may not be achieved. Due to these reasons, a profound understanding of the die filling process in order to develop an accurate model to predetermine the filling result is required.

The die filling on rotary presses is a complex process due to the different machine geometries and parameters which have an impact on the die filling result: the design of the fill cam, the filling angle, the dosing height, the number and design of paddles in the feed frame, the turret and paddle speed as well as powder properties of the respective formulation have to be taken into account. Generally, two different mechanisms of die filling can be distinguished: gravity and suction filling. While the lower punch is already at a low position during gravity filling, it is pulled down under the powder bed during suction filling, resulting in a negative differential pressure, which enhances the powder flow into the die [10].

Several studies focused on the understanding of the filling process on linear filling devices, consisting of a moving feed shoe and a stationary die, which simplifies the process and reduces the impact parameters compared with a rotary press (e.g. [10–16]). The experimental setup with transparent die walls enables the observation of different flow patterns during gravity filling depending on material properties: nose flow, bulk flow and intermittent flow [10]. Following, the setup was used to characterize flowability of pharmaceutical excipients [11], while beside gravity filling also suction filling [12] as well as filling under vacuum was studied [13]. Additionally, models were developed to describe an apparent mass flow from the feed shoe into the static die. The critical shoe velocity, describing the maximum shoe velocity which allows complete filling, was used to evaluate the filling success on the linear filling devices. Jackson et al. presented a high impact of the suction filling mechanism on the filling efficiency compared to results obtained by gravity filling [12]. Schneider et al. found higher powder weights delivered into the die for gravity filling under vacuum, which the authors explained by an escaping air flow during gravity filling which counteracts the powder flow. Differences in maximum shoe velocities were observed depending on the powder properties of the investigated materials, whereby the permeability was discussed to have a large impact [13]. Following, the influence of air flow on the powder flow during gravity filling was investigated by using discrete element methods (DEM) and computational fluid dynamics (CFD). Guo et al. presented simulations, which display air velocities and resulting particle movement during gravity filling from a stationary shoe into a stationary die [17]. In detail, they found a normalized mass flow from the static feed frame to be proportional to the Archimedes number multiplied by the solid-air density ratio in a log-log plot, i.e. the relationship can be described by a power function. For a moving feed shoe, a power-law relationship was developed by Schneider et al. to describe the mass delivered into the die in dependence on the shoe velocity under gravitational forces in air as well as under vacuum. Nevertheless, the application of the model on a rotary press to predict a maximum production rate failed by underestimating the maximum turret speed by half [13]. The authors identified the neglected impact of suction pressure as main reason for the unexpected high powder weight filled on a rotary press. A comparison with results obtained by gravity filling on a rotary press was not performed. In order to cope with the additional forces during suction filling, Baserinia et al. extended the model based on suction pressure measurements on a die in a linear filling device [14]. Nevertheless, the application of the model on rotary presses is still challenging as the comparability between a linear filling device with a moving shoe and a rotary press with moving dies and a paddle feeder with minimum one wheel is limited.

The impact of a moving die on a die table with a static feed shoe without a paddle compared with the linear filling device was studied by Zhong et al., who found lower filling results for a poorly flowing powder on the system with moving dies. The results of the study emphasize the challenges of adapting the models developed on a linear device to a rotary press. Zakhvatayeva et al. used the experimental setup with a rotating die table in combination with the static feed shoe and investigated correlations between its critical velocity and measureable powder properties [18]. Among other measurements, they conducted pressure drop experiments to derive information about the powder permeability. An increase in pressure drop results in a strongly decreasing critical velocity at low pressure values, whereas above 1.5 mbar the effect on the critical velocity was low. A lower pressure drop can be correlated with a high permeability.

Nevertheless, the transferability of the developed models and findings is limited as the experimental setups do not take all impact factors, especially the paddle movement inside the feed frame, into account. Therefore, building a profound understanding of the die filling process directly on rotary tablet presses is essential for the development of a model to describe and predict the filling success in dependence on the process parameters and machine settings on one hand and the material properties on the other hand. As the filling process gets even more complex for suction filling due to additional forces besides the gravitational force, as well as for a double-wheel feed frame, the setting was reasonably and systematically limited to applying gravity filling with a feed frame with one paddle wheel. Thus, the aim of this study is to understand the gravity filling process on a pilot-scale rotary press in depth and, based on this, develop a model to predict the filling result.

2. Material and methods

2.1. Materials

Four different pharmaceutical excipients were used as model materials: microcrystalline cellulose (MCC, Vivapur® 200, JRS Pharma, Rosenberg, Germany), tricalcium citrate tetrahydrate (TCC, Tricalcium Citrate, Jungbunzlauer Ladenburg GmbH, Ladenburg, Germany), anhydrous dicalcium phosphate (DCP-A, DI-CAFOS® A150, Chemische Fabrik Budenheim KG, Budenheim, Germany) and dicalium phosphate dihydrate (DCP-D, DI-CAFOS® D160, Chemische Fabrik Budenheim, Budenheim, Germany). The powders have similar particle size

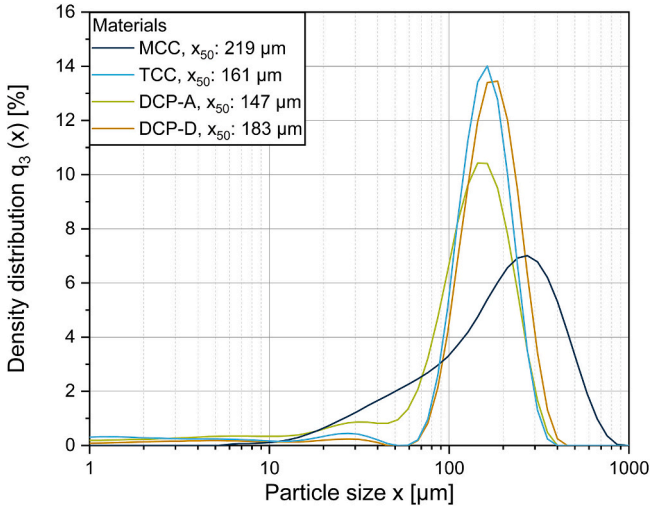


Fig. 1. Particle size distribution of the excipients.

distributions and comparable median particle sizes x_{50}, only MCC has a wider distribution with coarser particles (Fig. 1). Scanning electron microscope (SEM) images are shown in Fig. 2. The bulk, tapped and solids density varies for the four materials, while the highest bulk and tapped density can be found for DCP-D (Table 1.). DCP-A has the highest solids density.

Table 1
Bulk, tapped and solids densities of the excipients used.

Material	Bulk density ρ_b [g/cm^3]	Tapped density ρ_t [g/cm^3]	Solids density ρ_s [g/cm^3]
MCC	0.397 ± 0.002	0.498 ± 0.002	1.564 ± 0.036
TCC	0.579 ± 0.002	0.652 ± 0.003	1.959 ± 0.006
DCP-A	0.716 ± 0.011	0.839 ± 0.008	2.824 ± 0.127
DCP-D	0.900 ± 0.004	0.952 ± 0.001	2.322 ± 0.002

2.2. Methods

2.2.1. Powder characterization

Particle size distributions were measured by means of dry laser diffraction (MasterSizer 3000, Malvern, United Kingdom). Three measurements at a dispersion pressure of 0.5 bar were conducted and the mean values were calculated. Solids densities of the materials were characterized by helium pycnometry (Ultrapyc 1200e, Quantachrome Instruments, Boynton Beach, USA). Bulk and tapped density were determined according to the Ph. Eur. 9.3 2.9.34 using a 100 ml graduated cylinder, tapped by a volumetric analyzer (Erich Tschacher Laboratoriumsbedarf, Bielefeld, Germany). Three measurements were conducted and values were averaged. Permeability measurements were carried out with the powder rheometer MCR 302 (Anton Paar, Graz, Austria) under a consolidation stress of 1 kPa with an air-permeable punch. In order to ensure unconsolidated conditions and homogeneously packed particles before consolidation, 80 ml of powder was filled into the powder cell and fluidized for 200 s using a volume flow of 5 l/min, which was sufficient for all powders to obtain a completely fluidized powder bed. A two-blade stirrer was installed, which rotated with 5 rpm. After fluidization, a compression tool with air-permeable punch was installed which moved downwards until a normal stress of 1 kPa was exerted onto the powder bed. Following, two different air volume flows were introduced into the powder bed from the bottom, 0.5 l/min and 1 l/min respectively. For calculation of the permeability, Darcy's law was used by relating the applied volume flows and respective differential pressures (Eq. (11)).

2.2.2. Die filling experiments

The die filling experiments were carried out on a pilot-scale rotary press XL 100 (Korsch AG, Berlin, Germany). A mixed-rotor, which can be either used for Euro-D or Euro-B tools, was equipped with four sets of round, flat-faced 9 mm Euro-D tools. For powder feeding, two different feed frames with one paddle, respectively, were used (Fig. 3 a). Fig. 3 a shows an illustration of the counter-clockwise rotating turret (black circle) which travels below the feed frame. Inside the feed frame, the area in which the powder is agitated during tableting experiments is indicated in light blue. Due to constructional reasons, feed frame A exhibits a metal part in front of the filling area which seals the dies from air. Feed frame B additionally possesses a extended opening in front of the normal filling area. For the experiments, this additional opening was equipped with an air-permeable frit (marked in gray) to compensate possible differential pressures resulting from punch pull-down in that area. A fill cam which is dedicated for filling with an external lubrication unit was used (marked in red in Fig. 3 a). As a consequence, the lower punch is already pulled down before entering the filling area of the feed frame, illustrated in Fig. 3 b. Thus, no suction pressure is expected there. In its original purpose, this fill cam enables the coverage of the die wall and lower punch surface with a lubricant before powder is filled into the dies. In this study, this respective fill cam is used in order to study gravity filling on the pilot-scale rotary press. Fig. 3 b displays the position of the lower punch surface below the turret surface. Here, a small overlap of the downward movement and the powder bed above, indicated in light blue, is still observable.

Tablets were produced at different process parameter combinations to characterize the filling behavior. The turret speed was varied between 20 and 60 rpm in steps of 10 rpm, while the paddle speed was varied in seven steps (5, 10, 20, 30, 40, 50, 60 rpm). Filling height (16 mm) and dosing height (15.5 mm) were kept constant. Ten Tablets were collected after 30 s and automatically weighed by the MultiCheck 6 tablet tester (Erweka, Langen, Germany). Mean values and standard deviations were calculated. All experiments were carried out in triplicate. The developed linear model equation (see Sec. 3.3) was fitted to each of the three experiments per material individually. The fit parameters, the slope, called rotation-specific fill volume, V_r and the axis intercept, called reference volume flow, $\dot{V}_0$ were averaged in the following and the respective values used to calculate the critical paddle speed derived from experimental data.

2.2.3. Measurement of differential pressure

The differential pressure, which develops in the die during the pull-down of the lower punch is measured with the differential pressure sensor SDP810-500 Pa (Sensirion, Switzerland). The sensor can detect differential pressures of ±500 Pa against the ambient pressure with an accuracy of 3% of the measured value. In order to measure the resulting differential pressure in the die, which has an impact on the powder flow from the feed frame, experiments with an empty feed frame were conducted. The filling area above the rotating dies inside the feed frame was isolated from the environment by covering it with tape (Fig. 4). A silicone tube was used to connect the isolated volume between the tape-sealed bottom plate of the feed frame and the rotating turret with the differential pressure sensor. The other opening of the sensor was left open to enable a measurement against ambient conditions.

A clear signal was observed by using only one lower punch while three blind dies replace the other tools. The resulting differential pressure was measured every 1 ms for the five turret speeds used for tableting experiments. Every measurement was run for 10–20 s. The data was processed by using a moving average filter which used 21 values. In order to enable the comparison of the different differential pressure measurements, data points which cover no suction event were averaged

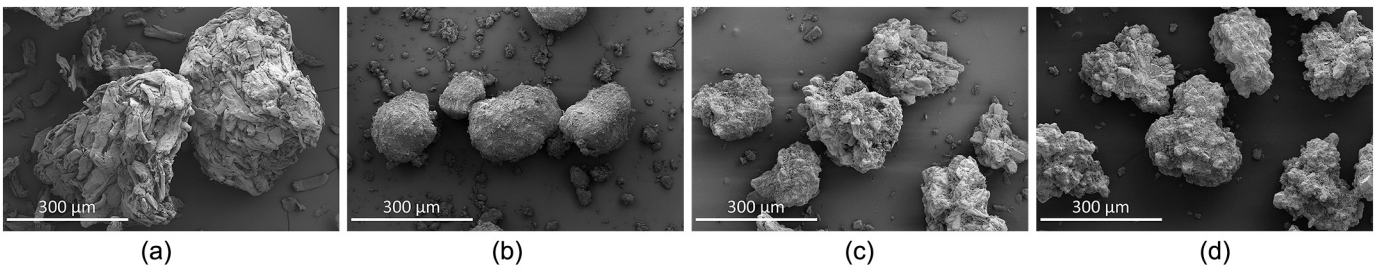


Fig. 2. SEM images of the excipients: MCC (a), TCC (b), DCP-A (c), DCP-D (d).

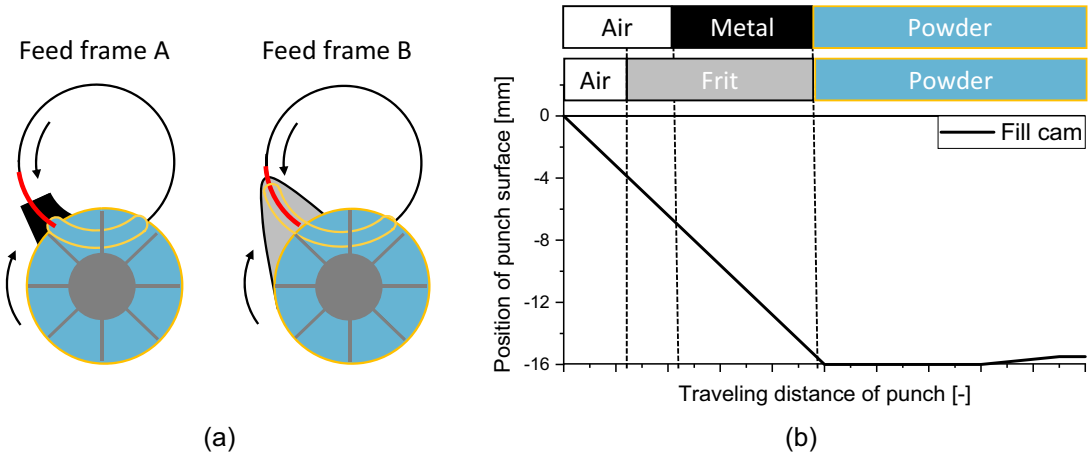


Fig. 3. Schematic image of the feed frames used including the die trace on the turret (black circle) (a); Illustration of the position of the punch surface in relation to the feed frame above (b).

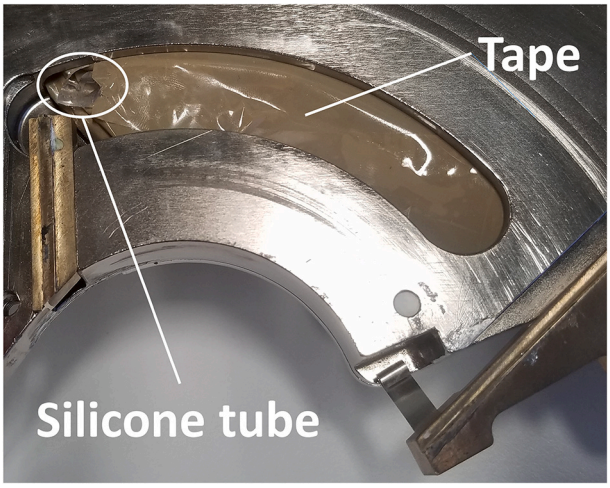


Fig. 4. Photo of the prepared feed frame for suction measurements from underneath.

to calculate a baseline offset. The respective data set was then corrected by applying the baseline offset. The data points representing the filling were identified and the minimal values which correspond to the highest differential pressure were extracted using a Matlab script to compare different turret speeds.

3. Results and discussion

3.1. Measurement of suction pressure

The aim of this study was to investigate the gravity filling on the pilot-scale rotary press XL 100. During gravity filling, the lower punch is already in a low position as it enters the powder-filled area of the feed frame. The position at which the lower punch crosses under the powder corresponds to the filling height. The configuration of feed frame and fill cam of the applied rotary tablet press prevents a complete pull-down in front of the feed frame. However, due to the possibility to apply an external lubrication, the remaining overlap of pull-down and feed frame is small (Fig. 3 b). Nevertheless, the arising suction pressure was investigated with a differential pressure sensor (Sec. 2.2.3). Depending on the used feed frame, different pressure profiles during a single compaction cycle were measured and exemplarily displayed for three turret speeds (Fig. 5 a). The differential pressure becomes increasingly negative during the pull-down, has a minimum and subsequently rises above zero. This change from suction to overpressure was also observed on a linear filling device by Baserinia et al. [14]. A second increase of the differential pressure can be correlated to dosing, whereby the lower punch moves upwards again. To allow a statistically meaningful comparison of the suction pressure profiles, minimal pressure values were evaluated for five turret speeds (comp. Sec. 2.2.3., Fig. 5 b).

In the measurement data (Fig. 5 a) as well as in the processed minimum pressure data (Fig. 5 b), an increasing suction pressure with rising turret speed is observable if the external lubrication fill cam and feed frame A are combined. Therefore, this setting is not able to represent the filling mechanism ‘gravity filling’. In contrast, the application of feed frame B, which uses an air permeable frit at the inlet of the filling area (Fig. 3 a), results in very small suction pressures for the entire range of applied turret speeds. These low values are caused by the possible pressure compensation by an airflow through the frit before the die enters the filling area, in which the silicone tube was installed and the measurement took place. Thus, this setting better represents gravity filling and is consequently used for die filling experiments in this study.

3.2. Die filling experiments

Tablets must achieve a predefined and consistent weight to ensure uniformity of the dose of active pharmaceutical ingredients (APIs). The expectable weight for completely filled dies can be calculated by using the dosing volume and bulk density. Following, process parameters which lead to completely filled dies have to be identified in dependence on the powder properties. In order to gain a deeper process understanding, die filling experiments with four excipients MCC, TCC, DCP-A and DCP-D were conducted using the rotary press XL 100 in pilot-scale. For each powder, the tablet weight increases with increasing paddle speed until it reaches a constant value, which indicates complete filling. With increasing turret speed, the tablet weight decreases for a given paddle speed (exemplarily shown for TCC and DCP-D in Fig. 6 a, b). Those findings are in good agreement with results found in the literature [2,4,5,19]. Nevertheless, as the graphs show, a comparison of different materials regarding the filling ratio is not possible. Tablet weights differ strongly due to the different densities of the materials. Here, a normalization is necessary which can be conducted by calculating the filling yield [5]. The calculation of the filling yield φ_{bulk} is based on the bulk density ρ_b, the tablet weight m_T, the die radius r_{die} and the dosing height h_D (Eq. (1)). Following, this value enables the evaluation of the filling result and especially the comparison of different materials, as exemplarily presented for TCC and DCP-D (Fig. 6 c, d). Complete die filling is indicated by a value of 1. Nevertheless, plotting the filling yield for

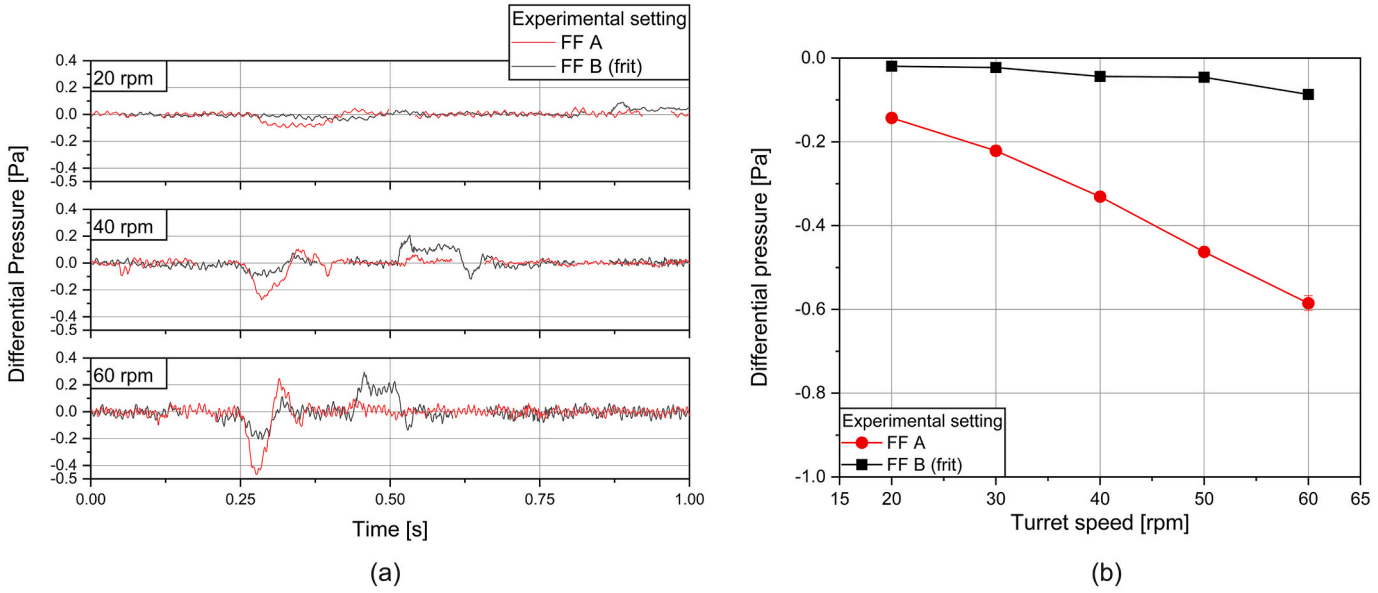


Fig. 5. Differential pressure over time for turret speeds of 20, 40 and 60 rpm for different experimental settings (a); Mean values of the minimal differential pressure over turret speed for the two feed frames (b).

different turret speeds for one material over the paddle speed does not result in one representative correlation, i. e. the filling yield still depends on the turret speed. To converge the curves for different turret speeds over the paddle speed and accordingly facilitate the development of a model equation, the turret speed must accordingly be taken into account.

$$\varphi_{bulk} = \frac{m_T}{\pi \bullet r_{die}^2 \bullet h_D \bullet \rho_b} \tag{1}$$

3.3. Model development

Besides the normalization of the tablet weight by the filling yield, further considerations are necessary to enable the development of a model to describe die filling in dependence on the process parameters and machine settings as well as on the material properties. Therefore, the volume flow into the dies was utilized to create a normalized value which is comparable for different materials. In order to calculate the achieved volume flow, the filling time has to be determined. It describes the time during which a die is below the feed frame and thus, can be filled with powder. The filling time can be calculated by dividing the opening angle of the feed frame α_{fill} by the angular velocity of the turret speed n_t:

$$t_{fill} = \frac{\alpha_{fill}}{n_t \bullet 360°} \tag{2}$$

Subsequently, the volume flow into the dies is calculated:

$$\dot{V} = \frac{m_T}{\rho_b \bullet t_{fill}} \tag{3}$$

with tablet weight m_T, bulk density ρ_b and the filling time t_{fill}. The calculation of the volume flow by the filling time is based on the assumption that changes in powder flow over the respective filling time are negligible. Although filling times are very short with values between 0.2 and 0.4 s, particles inside the feed frame are in constant movement due to the paddle rotation which should lead to constant conditions and thus, a constant flow. Nevertheless, air flow out of or into the dies might has an additional impact. For the investigated powders, this assumption is valid as the volume flow for different turret speeds, and thus different filling times, are overlapping before a plateau is reached dependent on the turret speed (Fig. 7). In the future, it is interesting to investigate if this assumption remains valid for formulations with poor flowability.

In addition to the turret speed n_t, the counteracting paddle speed n_p has a considerable impact on the volume flow. For all investigated excipients, a linear relationship between volume flow $\dot{V}$ and paddle speed n_p was found until the volume flow reaches a constant value which depends on the turret speed (Fig. 7). These constant values can be explained by complete die filling which restricts further volume flow into the dies. The volume flow for complete filling $\dot{V}_{comp}$, which is necessary to enable complete die filling, can be calculated by considering the dosing volume V_D and the filling time t_{fill}:

$$\dot{V}_{comp} = \frac{\pi \bullet r_{die}^2 \bullet h_D}{\alpha_{fill}/(n_t \bullet 360°)} = \frac{V_D}{t_{fill}} \tag{4}$$

The experimentally found constant values of the volume flow match these theoretical values for each respective turret speed (Fig. 7). Therefore, in the experiments, the dies are filled completely while no densification inside the dies occurs, which would be indicated by an exceedance of the calculated volume flow for complete filling $\dot{V}_{comp}$. In a previous study, densification was observed for MCC, depending on the overfill ratio, and in a minor extend for DCP-A, depending on the overfill ratio and strongly on the paddle speed [5]. However, as the overfill ratio in this study is low with 0.03, no densification is expected.

Based on these findings, regarding the description of the filling process, only data points on the linearly increasing section of the graphs correlate with incomplete die filling. Here, data points of different turret speeds converge which facilitates the development of a model describing the delivered volume flow in dependence on the paddle speed.

For the model, a linear equation is selected to describe the impact of paddle speed on the volume flow as it represents the data well (Fig. 8, Eq. (5)).

$$\dot{V} = V_r \bullet n_p + \dot{V}_0 \tag{5}$$

The slope, called rotation-specific fill volume V_r, describes the correlation between the volume flow and the paddle speed. The second parameter of the model, the axis intercept, describes the theoretical volume flow without any agitation of the powder by the paddle wheel. Accordingly, this parameter is called reference volume flow $\dot{V}_0$. In order to predict the filling results, the model has to be fitted on the obtained data sets to extract V_r and $\dot{V}_0$ for each material which can be deduced

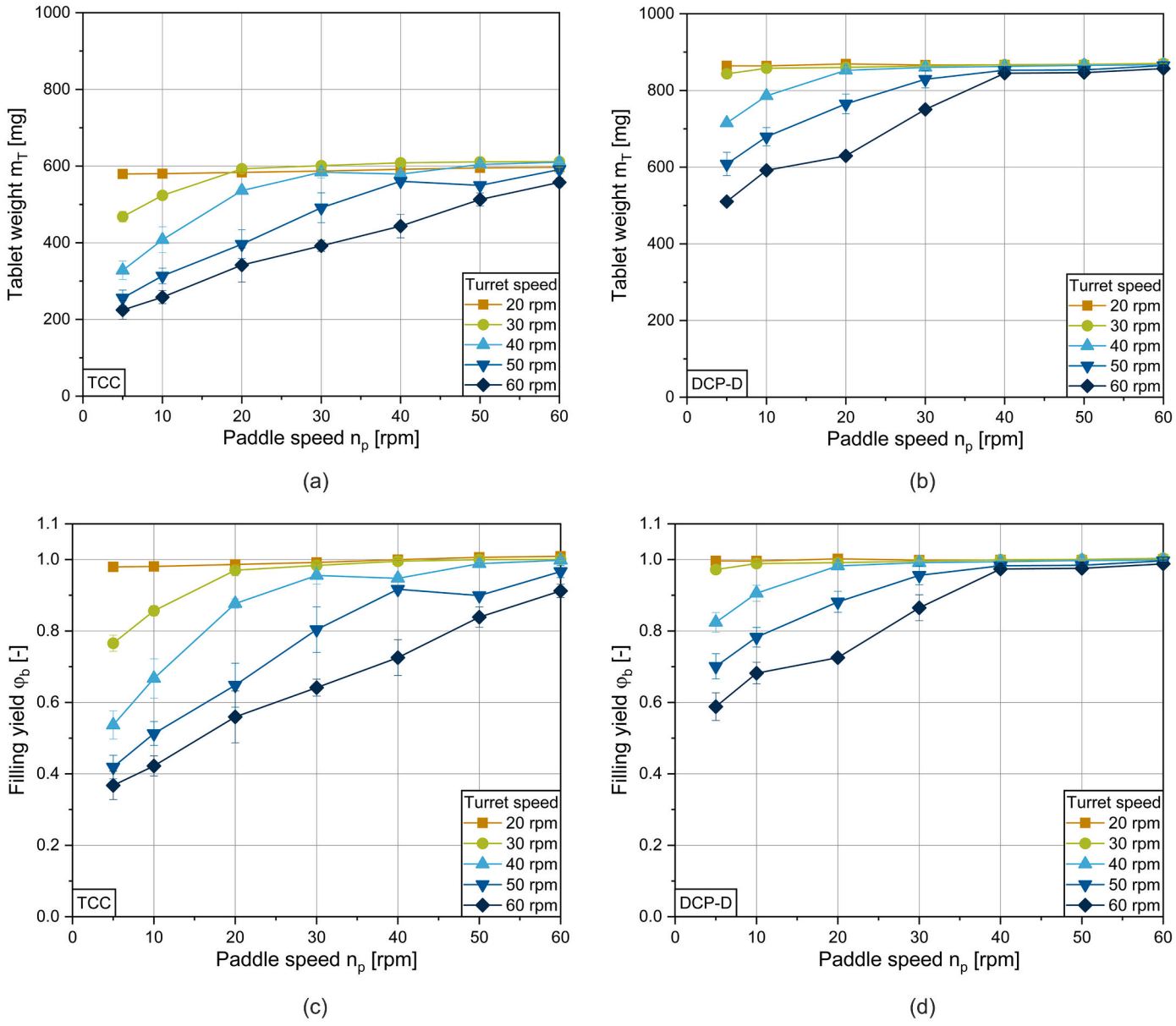


Fig. 6. Tablet weight for TCC (a) and DCP-D (b); Filling yield for TCC (c) and DCP-D (d).

from Fig. 7.

The descriptive model allows to determine the volume flow into the die at any process parameter combination of turret and paddle speed for the investigated material. Using the volume flow, an expected mean tablet weight can be calculated within the range of not completely filled dies. For this, the volume flow $\dot{V}$, based on the a predefined paddle speed n_p, is multiplied by the filling time t_{fill}, which depends on the turret speed n_t, and by the bulk density ρ_b of the material of interest:

$$m_T = t_{fill}(n_t) \bullet \dot{V}(n_p) \bullet \rho_b \tag{6}$$

As the model can be set up by only two data points in extreme, as it is a linear function, only two combinations of paddle and turret speed in the region of incomplete filling need to be studied to be able to calculate any resulting tablet weight.

However, in most cases, the tablet weight is one of the target parameters as it is closely linked to the API dose for a defined formulation. Additionally, a high turret speed corresponding to a specific production rate might be predefined based on the demand of the market. With these parameters, a minimal required paddle speed can be determined using the model equation to ensure complete filling on one hand and low shear stresses, which could limit the amount of lubricant dispersion, abrasion and demixing, on the other hand. For this purpose, the paddle speed at the intersection of the volume flow for complete filling $\dot{V}_{comp}$ for the predefined turret speed and the apparent volume flow $\dot{V}$ in dependence on the paddle speed has to be calculated. Therefore, the two equations Eq. (4) and Eq. (5) have to be equalized (according to Eq. (7)) and solved for n_p:

$$\dot{V} = \dot{V}_{comp} \tag{7}$$

$$n_{p,crit} = \left(\frac{V_D \bullet n_t \bullet 360^\circ}{\alpha_{fill}} - \dot{V}_0 \right) \bullet \frac{1}{V_r} \tag{8}$$

The transformation into the critical paddle speed $n_{p,crit}$ represents a linear dependency on the dosing volume V_D and the turret speed n_t. Thus, for realizing higher fill volumes and higher production rates, the paddle speed has to be increased according to prior findings in this paper (Fig. 6) and previous studies [2,4,5,19]. Furthermore, the equation as function of the turret speed shows an inversely proportional correlation

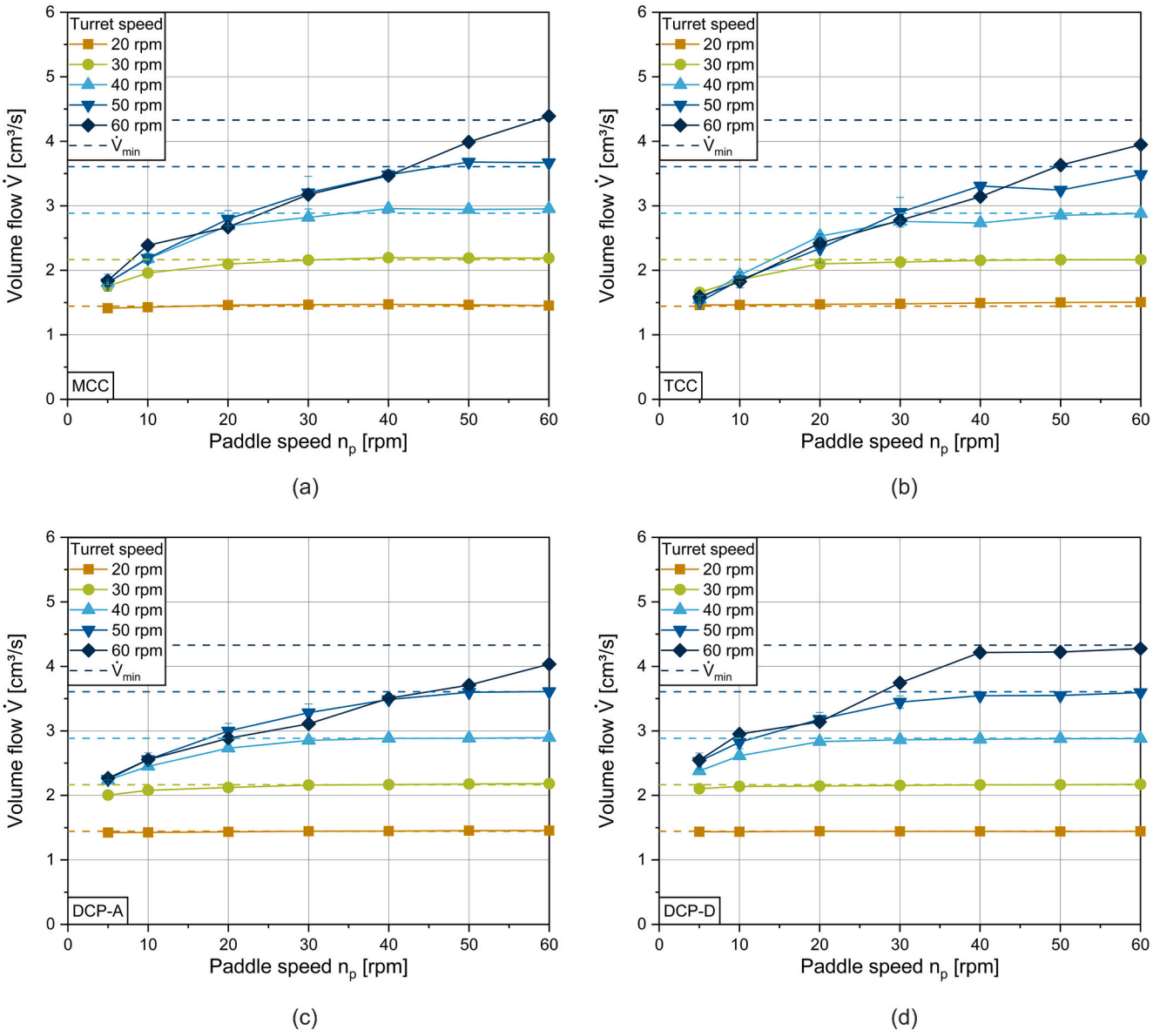


Fig. 7. Volume flow during die filling for the four excipients MCC (a), TCC (b), DCP-A (c) and DCP-D (d).

between the critical paddle speed and the open filling angle α_{fill}. Therefore, a higher filling angle could enable a further decrease of the paddle speed.

The presented model combines impacts of the machine geometry (i. e. filling angle) and process parameters. In addition, also the powder properties have an impact on the filling results. These properties are represented by the model parameters rotation-specific fill volume V_r and reference volume flow $\dot{V}_0$, while a high reference volume flow and a high rotation-specific fill volume is preferred. In order to understand the impact of measurable powder properties on the obtained model parameters V_r and $\dot{V}_0$, different measurements considering the flowability, bulk and tapped density and permeability were conducted and correlations were investigated (see Supplementary Material).

For the axis intercept a good linear correlation was found for the bulk density of the powders (Fig. 9 a). By using a linear fit function, an explicit determination of the axis intercept based on the bulk density is possible for the powders under investigation:

$$\dot{V}_0 = 2.263 \bullet \rho_b + 0.415 \tag{9}$$

Beside the bulk density, also the tapped density presents a good correlation. Nevertheless, as the bulk density is also necessary to calculate the volume flow (cf. Eq. (3)), it is convenient to stay with one parameter. The bulk density could be a good descriptor in this context, since it affects the gravitational forces acting while the bulk solid flows into the die: the higher the density, the higher the forces acting for a given volume. Nevertheless, it has to be emphasized that a correlation with high accuracy is only valid for the bulk density (Pearson R 0.94207) but not for the solid density (Pearson R 0.75625) (Fig. 10 a). Therefore, beside the gravitational force acting on single particles based on their solid density, further impact factors play an important role for the reference volume flow $\dot{V}_0$. As presented by several authors, air inside the die has to escape during gravity filling while particles are flowing into it [10,17,20]. Therefore, the air flow counteracts the downwards flow and thus has to be taken into account beside the gravitational forces. For powders with higher bulk porosity, this restraining effect might be smaller, whereas the results present the opposite as the axis intercept decreases by trend with increasing porosity (Fig. 10 b). Additionally, the strength of correlation is considerably weaker than for the bulk density (Pearson R − 0.69502 vs. 0.94207). Nevertheless, for this discussion, it has to be considered that only powders with similar particle size

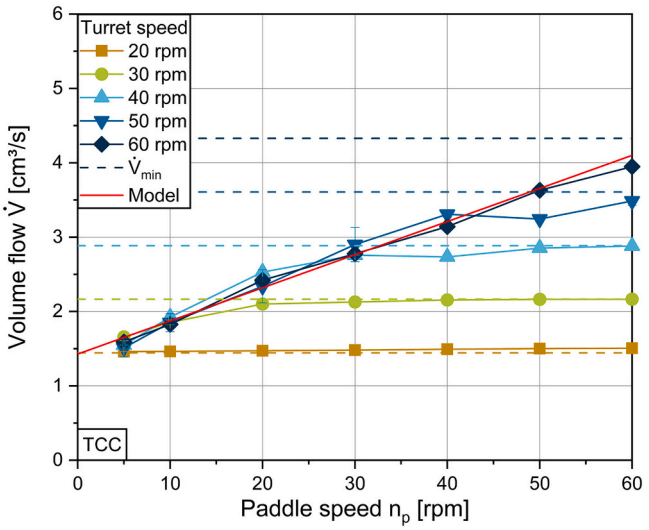


Fig. 8. Model development by applying a linear fit to data of not completely filled dies.

distributions were investigated. A clear impact of the particle size distribution, which is only slightly wider for MCC, with coarser and finer particles, might not be identified in this work. For future studies, the proposed model and the found correlations with powder properties have to be carefully extended to powders with varying particle size distribution.

In contrast to the reference volume flow, the rotation-specific volume describes the increase in delivered powder volume under increasingly dynamic conditions. In several studies, the positive impact of the paddle movement was presented but until now no model was developed to describe or predict it. For the slope, a linear dependency on the measured permeability K of a consolidated powder bed under stresses of 2 kPa was found (Fig. 10 b):

$$V_r = -1.59837 \bullet 10^{11} \bullet K + 3.564 \tag{10}$$

As already discussed for the axis intercept, the escaping air from the dies during gravity filling plays an important role. For a given pressure drop Δp, the air flow through a powder bed $\dot{V}_{air}$ is higher for high values of permeability K according to the Darcy-equation:

$$\dot{V}_{air} = \frac{K}{\eta} \bullet \frac{A}{L} \bullet \Delta p \tag{11}$$

with the dynamic viscosity η, the cross-sectional area A and the characteristic length L [21,22]. Contrarily to the expectations, the slope

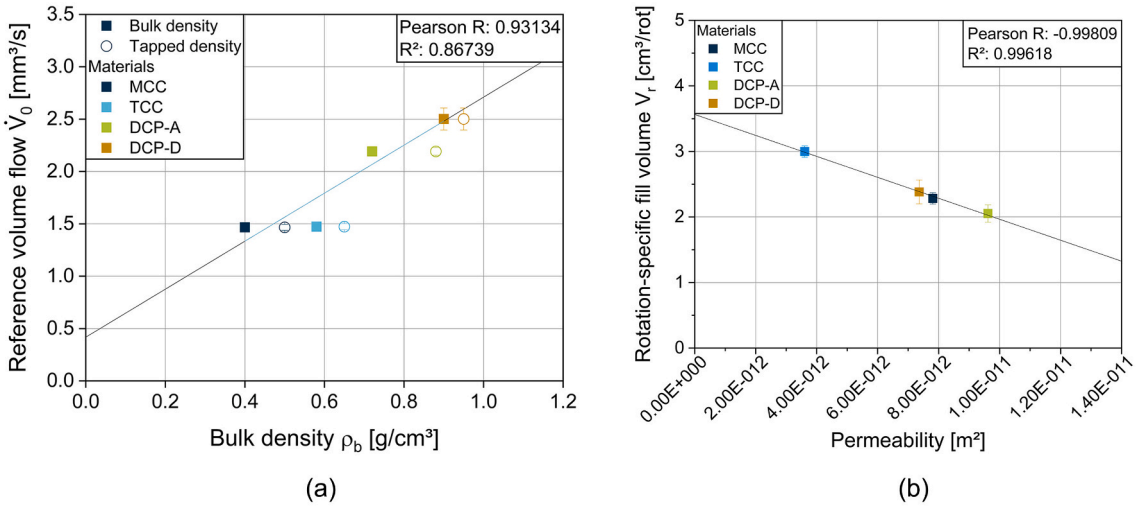


Fig. 9. Correlation for the axis intercept (reference volume flow) with the bulk density of the investigated powders (a) and for the slope (rotation-specific volume) with the permeability at a compression stress of 1 kPa (b).

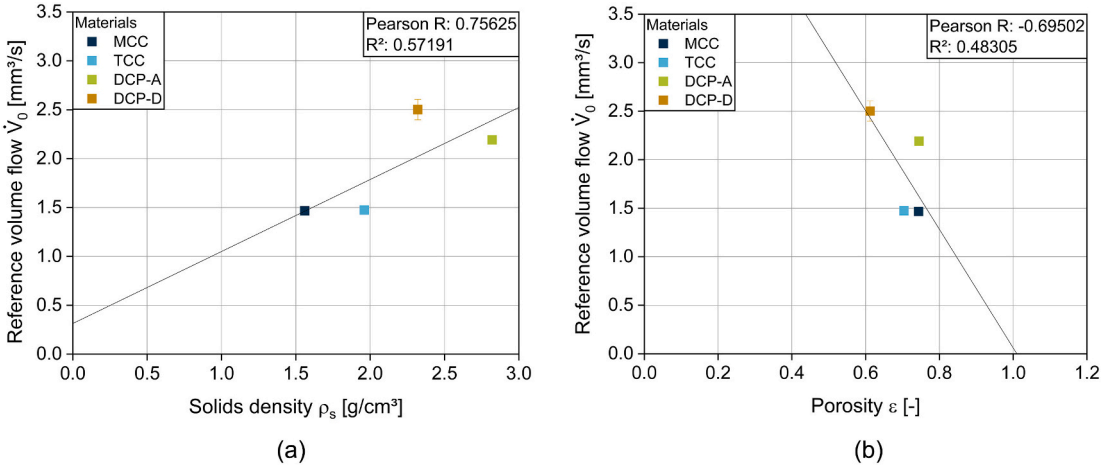


Fig. 10. Correlation for the axis intercept (reference volume flow) with the solid density of the investigated powders (a) and with their bulk porosity (b).

decreases with rising permeability, which means the powder volume flow increases less with rising paddle speed for powders with high permeability. This could be explained by the fluidization state inside the feed frame. Powders with low permeability present a low volume flow for small paddle speeds while the increase in rotation speed slightly fluidizes the powder bed leading to a higher apparent permeability and thus, a higher air volume flow escaping from the die. Powders which initially present a high permeability, do not benefit as much from a potentially increased permeability – perhaps, here, the extent of fluidization is even lower. Therefore, the slope decreases with increasing permeability. However, this has to be investigated in more detail in the future.

Using the correlations for the rotation-specific volume V_r and the reference volume flow $\dot{V}_0$, the model can be extended:

$$\dot{V} = \left(-1.59837 \bullet 10^{11} \bullet K + 3.564 \right) \bullet n_p + 2.263 \bullet \rho_b + 0.415 \quad (12)$$

The present model now allows to include not only the process parameters but also the powder properties to describe the filling success. Converting and applying the model to calculate the critical paddle speed $n_{p,crit}$, leads to the following equation:

$$n_{p,crit} = \left(\dot{V}_{min} - (2.263 \bullet \rho_b + 0.415) \right) \bullet \frac{1}{-1.59837 \bullet 10^{11} \bullet K + 3.564} \quad (13)$$

In order to evaluate the accuracy of the presented model, the predicted critical paddle speed $n_{p,crit}$ (Eq. (13)) required for complete die filling at turret speeds of 20, 30, 40, 50 and 60 rpm is compared with the critical paddle speed determined experimentally in the diagrams. In experimental determination, the intersection of the fit described above with the volume flow for complete filling $\dot{V}_{comp}$, according to Eq. (7) and Eq. (8), is sought. The parity plot of both values presents a very good agreement between experimentally determined and predicted values by using Eq. (13) (Fig. 11). The coefficient of determination (CoD/R^2), a statistical measure describing the quality of the linear regression, is very high with 0.983. Therefore, the proposed model shows a very good predictive power for the critical paddle speed for the investigated materials on the rotary press in pilot-scale. Only for TCC, a systematic deviation is observable, which can be traced back to the lower accuracy of the axis intercept by the bulk density for this material compared to the other three ones (Fig. 9 a).

4. Conclusion

In this study, a model was introduced to predict the filling results based on process parameters and material properties. Filling behavior of four pharmaceutical excipients with similar particle size distributions but varying solids and bulk densities were investigated on a rotary press in pilot-scale. For comparison, the volume flow into the dies was calculated by dividing the tablet weight by the filling time and bulk density. By this normalization, different materials are comparable. The volume flow as function of paddle speed for all excipients present an accumulation of data points on a linear curve, for all cases, where the die is not completely filled. Based on this finding, a model equation was developed to describe the volume flow in dependence on the paddle speed. Using this model, a critical paddle speed can be calculated by combining a volume flow for complete filling at a given turret speed with the model equation. Solving the equation for the paddle speed results in the minimum rotation frequency which leads to completely filled dies at a given turret speed.

In order to describe not only the dependency on process parameters and machine geometry, but also consider powder properties, correlations between the model parameters rotation-specific volume and reference volume flow with measured material characteristics were investigated. For the axis intercept, the so-called reference volume flow, a positive linear correlation with the bulk density was found. Additionally, the permeability presents a negative linear correlation with the slope, the so-called rotation-specific volume. In order to evaluate the capability of the model to predict the critical paddle speed by considering these relations, values were compared with experimentally derived ones. The parity plot presents a very good agreement.

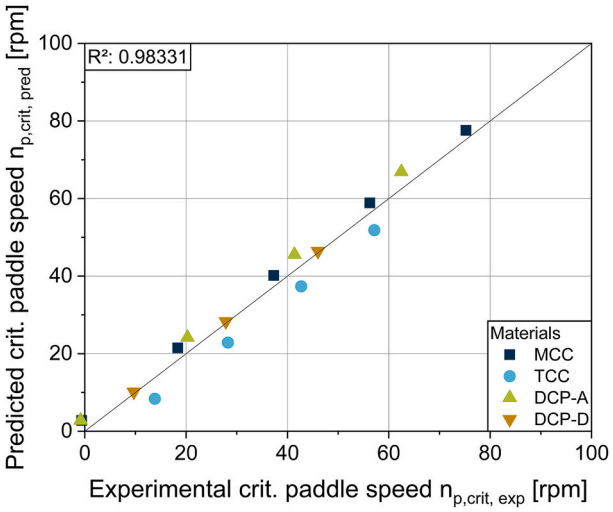


Fig. 11. Parity plot of the predicted critical paddle speed $n_{p,crit,pred}$ over the experimentally determined critical paddle speed $n_{p,crit,exp}$.

In future work, the applicability of the model to powders with varying particle size distributions have to be proved. Furthermore, the transfer to rotary presses in industrial scale has to be investigated and comparably precise models for suction filling should be developed.

CRediT authorship contribution statement

A.K. Schomberg: Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **A. Kwade:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing. **J.H. Finke:** Conceptualization, Investigation, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by Korsch AG. Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by Chemische Fabrik Budenheim KG. Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by Jungbunzlauer Ladenburg GmbH. Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by J Rettenmaier and Sons.

Data availability

Data will be made available on request.

Acknowledgements

This paper evolved of the research project “Simulation of distributed product structures in combined discrete and continuous production processes for solid, particulate products” which is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – 413141366. We acknowledge support by the German Research Foundation. Furthermore, the authors acknowledge and thank KORSCH AG for the support and the provision of the tablet rotary press XL 100. For

the provision of testing materials, the authors thank Chemische Fabrik Budenheim KG (DI-CAFOS® A150 and DI-CAFOS® D160), Jungbunzlauer Ladenburg GmbH (Tricalcium Citrate (TCC TB)) and JRS - J. Rettenmaier & Söhne GmbH + Co KG (Vivapur® 200). The authors thank Karl Vorländer for providing the SEM-images.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.powtec.2022.117998.

References

[1] B. van Snick, J. Holman, C. Cunningham, A. Kumar, J. Vercruysse, T. Beer, et al., Continuous direct compression as manufacturing platform for sustained release tablets, Int. J. Pharm. 519 (1–2) (2017) 390–407. Available from: URL: https://www.sciencedirect.com/science/article/pii/S0378517317300108.

[2] R. Méndez, F. Muzzio, C. Velazquez, Study of the effects of feed frames on powder blend properties during the filling of tablet press dies, Powder Technol. 200 (3) (2010) 105–116.

[3] H.P. Goh, P.W. Sia Heng, C.V. Liew, The effects of feed frame parameters and turret speed on Mini-tablet compression, J. Pharm. Sci. 108 (3) (2019) 1161–1171.

[4] E. Peeters, T. Beer, C. Vervaet, J.-P. Remon, Reduction of tablet weight variability by optimizing paddle speed in the forced feeder of a high-speed rotary tablet press, Drug Dev. Ind. Pharm. 41 (4) (2015) 530–539.

[5] A.K. Schomberg, A. Kwade, J.H. Finke, The challenge of die filling in rotary presses—a systematic study of material properties and process parameters, Pharmaceutics 12 (3) (2020).

[6] R. Mendez, C. Velazquez, F.J. Muzzio, Effect of feed frame design and operating parameters on powder attrition, particle breakage, and powder properties, Powder Technol. 229 (2012) 253–260.

[7] A.S. Narang, V.M. Rao, H. Guo, J. Lu, D.S. Desai, Effect of force feeder on tablet strength during compression, Int. J. Pharm. 401 (1–2) (2010) 7–15. Available from: URL: https://doi.org/10.1016/j.ijpharm.2010.08.027.

[8] W. Grymonpré, V. Vanhoorne, B. van Snick, B. Blahova Prudilova, F. Detobel, J. P. Remon, et al., Optimizing feed frame design and tableting process parameters to increase die-filling uniformity on a high-speed rotary tablet press, Int. J. Pharm. 548 (1) (2018) 54–61.

[9] D. Puckhaber, J.H. Finke, S. David, M. Serratoni, U. Zafar, E. John, et al., Prediction of the impact of lubrication on tablet compactibility, Int. J. Pharm. 617 (2022), 121557.

[10] C.-Y. Wu, L. Dihoru, A.C. Cocks, The flow of powder into simple and stepped dies, Powder Technol. 134 (1–2) (2003) 24–39.

[11] I.C. Sinka, L.C.R. Schneider, A.C.F. Cocks, Measurement of the flow properties of powders with special reference to die fill, Int. J. Pharm. 280 (1–2) (2004) 27–38.

[12] S. Jackson, I.C. Sinka, A.C.F. Cocks, The effect of suction during die fill on a rotary tablet press, Eur. J. Pharm. Biopharm. 65 (2007) 253–256.

[13] L. Schneider, I.C. Sinka, A. Cocks, Characterisation of the flow behaviour of pharmaceutical powders using a model die–shoe filling system, Powder Technol. 173 (1) (2007) 59–71. Available from: URL: https://doi.org/10.1016/j.powtec.2006.11.015.

[14] R. Baserinia, I.C. Sinka, Powder die filling under gravity and suction fill mechanisms, Int. J. Pharm. 563 (2019) 135–155.

[15] A. Zakhvatayeva, C. Hare, C.Y. Wu, Suction filling of pharmaceutical powders, Powder Technol. 355 (2019) 438–448.

[16] W.-Z. Zhong, A. Zakhvatayeva, L. Zhang, C.-Y. Wu, Powder flow during linear and rotary die filling, Int. J. Pharm. 602 (2021), 120654.

[17] Y. Guo, K.D. Kafui, C.Y. Wu, C. Thornton, A coupled DEM/CFD analysis of the effect of air on powder flow during die filling, AICHE J. 55 (1) (2009) 49–62.

[18] A. Zakhvatayeva, W. Zhong, H.A. Makroo, C. Hare, C.Y. Wu, An experimental study of die filling of pharmaceutical powders using a rotary die filling system, Int. J. Pharm. 553 (1–2) (2018) 84–96.

[19] Y. Yaginuma, Y. Ozeki, M. Kakizawa, S.-I. Gomi, Y. Watanabe, Effects of powder flowability on die-fill properties in rotary compression, J. Drug Deliv. Sci. Technol. 17 (3) (2007) 205–210.

[20] C.-Y. Wu, Y. Guo, Numerical modelling of suction filling using DEM/CFD, Chem. Eng. Sci. 73 (2012) 231–238.

[21] H. Darcy, Les fontaines publiques de la ville de Dijon: Exposition et application, Victor Dalmont, 1856.

[22] M. Muskat, The flow of homogeneous fluids through porous media, Soil Sci. 46 (2) (1938) 169. Available from: URL: https://journals.lww.com/soilsci/citation/1938/08000/the_flow_of_homogeneous_fluids_through_porous.8.aspx.